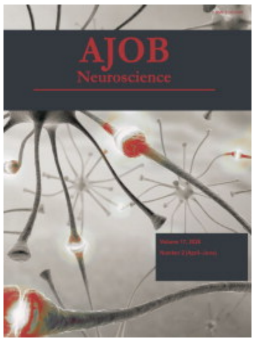

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Research on Brain Organoids Should Prioritize Questions of Agency, Not Consciousness

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Research on Brain Organoids Should Prioritize Questions of Agency, Not Consciousness

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The target article by Van Gyseghem et al. (2026) surveys discussions of consciousness in papers involving human brain organoids (HBOs). They rightly point out that there are many divergent concepts and theories of consciousness in the philosophical and scientific literatures, as well as a longstanding uncertainty about how to empirically detect consciousness or its most reliable markers, especially in liminal cases. Arguably, HBOs are one of these hard cases, despite presenting as a tractable alternative to the complexities of animal brains. Although they originate from the stem cells of human embryos that would normally mature into conscious humans, and although they exhibit some neural correlates of consciousness (NCCs) such as high-frequency oscillations (Trujilo et al., 2019), HBOs are unlikely to be conscious in the absence of embodiment (Croxford and Bayne 2024).

Using text-mining, Van Gyseghem et al. performed a meta-analysis of the HBO literature, employing methods outlined by Strech and Sofaer (2012) that "take into account the specific conceptual and practical challenges of empirical bioethics" (Van Gyseghem et al. 2026, 79). They identify six major themes in discussions of consciousness within the HBO literature to clarify which concepts of consciousness could plausibly be attributed to HBOs. Unsurprisingly, among the topics discussed are the predominant rival theories of consciousness, various methods that are purported to provide evidence of consciousness, the nature of such evidence in humans and other creatures, and the diversity of terminology used in discussions of consciousness. In a move that doesn't follow obviously from the data, the authors, noting the lack of consensus in the field, advocate for a pragmatic approach to consciousness in HBOs, and they ultimately suggest that it may be more productive to study organoid intelligence rather than consciousness. Not coincidentally, some of the most popular and highly cited experimental investigations of higher cognitive functions *in vitro* have conflated intelligence and consciousness (see, e.g. Kagan et al. 2022).

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While studying HBO intelligence may well be worthwhile, we suggest that studying agency would be a more productive research direction, particularly if one is interested in consciousness. We argue that agency is likely a biological prerequisite of consciousness, that it too has moral implications, and that it is more amenable to empirical investigation than is consciousness. Understanding agency and the role that organoids play in hybrid systems that integrate neural tissue with sensors and effectors could ultimately play a greater role both in advancing consciousness studies, and in determining the moral stakes of organoid research.

AGENCY AS A PREREQUISITE FOR CONSCIOUSNESS

The recent development of deep artificial neural networks has produced systems that strike many as intelligent; however, it is doubtful that those systems are conscious. Despite the lack of a touchstone to verify this claim, on many metrics, whether they be architectural, dynamic, metacognitive, or other, systems like large language models (LLMs) lack the embodiment, recurrent dynamics, complexity, and integrated behavioral profiles that characterize systems we are more certain display signatures of consciousness (Taylor et al. 2022). Thus, it is plausible that intelligence can exist apart from consciousness. In contrast, it is likely that all systems that have consciousness are also agentive: they are bounded systems embodied and embedded in the world, with interests or goals, and abilities to pursue them. Since consciousness evolved, it almost certainly has functions that confer fitness value, perhaps endowing agents with valenced states that are intrinsically motivating, better ways of representing and trafficking goal-relevant information, more flexible context-dependent processing, and in some cases self-models that anticipate and interact with other complex systems (see, e.g., the *Phil Trans Royal Society B* issue edited by Fitch, Roskies and Allen, 2025). What would the function of consciousness be in a system that had neither sensors nor effectors, that was merely passively awash in the surrounding environment, with neither goals nor interests?

EXPLORING AGENCY WITH HBOS

HBOs are embryonic-like models of animal brains, simplified in form and function, that can be instructed to recapitulate features of specific brain regions, such as the cortex or thalamus. These, in turn, can be combined as "assembloids" to model brain circuits, or coupled to sensors and effectors, such as retinal or muscular tissues – representing minimal forms of embodiment. While most HBOs are disembodied tools for disease modeling and drug discovery, several investigators have explored the role of closed-loop electrophysiology as a means of studying cognition in a dish. For example, building on the seminal works of Potter and colleagues (Demarse et al. 2001), and later Kagan et al. (2022), demonstrating that neural cells on microelectrode arrays (MEAs) spontaneously solve problems when receiving closed-loop feedback, Robbins et al. (2024) reported goal-directed learning in cortical HBOs tasked with balancing a pole on a cart in a virtual environment.

Van Gyseghem et al. (2026) acknowledge these lines of inquiry as important directions for consciousness and intelligence research; however, most investigators in the space agree that terms implying phenomenal experience such as "sentience" have been prematurely applied to these systems. Instead, HBOs and other neural tissues on MEAs in closed feedback loops have been termed "synthetic biological intelligences" and explored as biocomputing tools. Still, the tasks that have been assayed to date are fundamentally agential tasks, characterized by goal-directed action, such as learning to play virtual games or solve puzzles. Interestingly, the type of feedback is critical: positive reinforcement with predictable and structured stimulation is conducive to task success relative to chaotic or random inputs. That is, when a system's actions are reliably met with predictable reinforcement, the system tends to repeat those actions. In a closed-loop system, the contingency of the feedback determines the goal state, whether it is a biological imperative (e.g., drive, instinct, homeostatic setpoint) or a programmed neural-machine interface.

In addition to arguably serving a deeper functional role as a foundation for consciousness, from a pragmatic viewpoint, studying agency makes sense. Typically, mental states are inferred from observing action. Whether an HBO is intelligent, conscious, creative, or emotional, without some kind of interaction with external environments, observers will remain unaware of the presence of a mind. Agency as a capacity to initiate action is therefore one of the most tractable readouts of an embodied HBO, minimizing interpretive heavy lifting to make cognitive claims. HBOs demonstrating minimal agency will provide a basis for designing further experiments to assess both the structure of their neural networks, and the causal chains that precede action.

MORAL IMPLICATIONS OF AGENCY

Much of the ethical discussion about HBOs revolves around whether or not they are or could become conscious. While there is good reason to think that sentience, the ability to experience valenced states, does have moral implications, and self-consciousness has even more dramatic implications, there is also good reason to think that moral considerations can apply independently of sentience. We routinely confer moral weight to the furtherance or prevention of an agent's goals or interests, sometimes even in the absence of sentience, for example when we go to trouble to fulfill the wishes of the dead. An autonomous agent that differentiates itself from its environment and reliably acts in ways that articulate its goals, even when it is presented with obstacles, is an agent for which the goals in some sense matter. It has interests, and interests ground moral consideration. On this view, pleasure or pain matter only insofar as it is in or against an agent's interests to experience it. As agential capacities grow, so may the degree to which the plans or projects matter to the agent, and the moral consideration that may be given to it by others. The moral consideration due to simple agents may be minimal indeed, but let us not overestimate the degree of moral consideration that recognition of sentience in practice guarantees: the routine killing and poor treatment of animals who are not simply sentient, but can and do feel pain and likely suffer, belies the fact that we often allow concerns about moral status to be overridden by pragmatic considerations, whether or not that is morally justifiable. Exploration of agency in HBOs, at least for the foreseeable future, may be a good testing ground for scientific as well as moral theorizing about HBOs, given the difficulties attending attributions of consciousness.

In summary, given the complexities and ambiguities of establishing consciousness in simple organisms, we advocate for research that employs embodied HBOs, including hybrid robots, that are potentially agentive in order to better understand the bounds and implications of agency.

AUTHOR CONTRIBUTIONS

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